

# Python AI Agent From Scratch

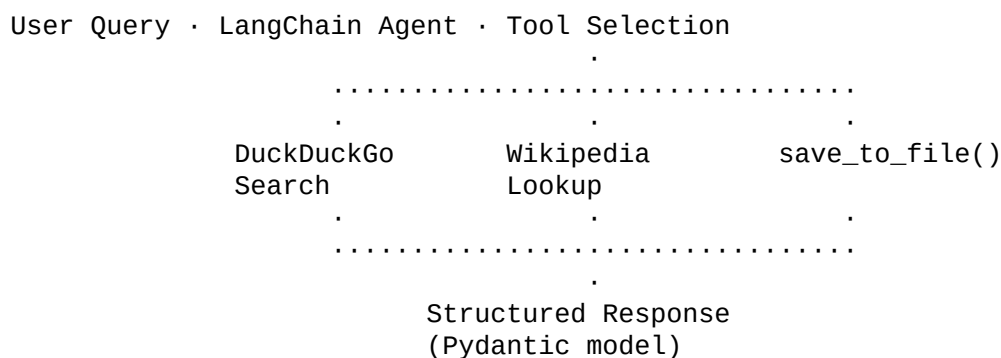
Build an AI agent from fundamental principles using **LangChain** with **Claude 3.5 Sonnet**, featuring **web search**, **Wikipedia lookup**, and **file saving** tools.

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## · Features

- **DuckDuckGo Search** · Web search for current information
  - **Wikipedia** · Encyclopedia lookup for knowledge queries
  - **File Saving** · Save research results to text files
  - **Claude 3.5 Sonnet** · Anthropic's powerful reasoning model
  - **Structured Output** · Pydantic-validated responses
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## .. Architecture



## · File Structure

| File                  | Purpose                                                  |
|-----------------------|----------------------------------------------------------|
| <code>main.py</code>  | Agent setup with Claude 3.5 Sonnet and structured output |
| <code>tools.py</code> | DuckDuckGo, Wikipedia, and file save tool definitions    |

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## · Getting Started

```
cd PythonAIAgentFromScratch
pip install -r requirements.txt
python main.py
```

## Environment Variables (.env)

```
ANTHROPIC_API_KEY=...
```

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## · Logic Flow

- 1 **Input** · User provides a research question

- 2 **Agent Reasoning** · Claude 3.5 Sonnet decides which tools to use
  - 3 **Tool Execution** · Web search, Wikipedia, or file operations
  - 4 **Structured Output** · Response validated against Pydantic schema
  - 5 **Display** · Clean, structured answer presented
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## · Dependencies

langchain, langchain-anthropic, duckduckgo-search, wikipedia, python-dotenv

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## · License

Educational project · use freely for learning and reference.